GREECE ECONOMIC CRISIS

Submitted as an Internal Assessment

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Motivation for Our Study

The Greek economic crisis illustrates the powerful connection between fiscal policy and global finance. Greece transformed from a thriving economy to facing significant debt challenges, providing invaluable lessons in governance and financial management. We delve into the factors that led to this crisis, including debt mismanagement, global influences, and the trials of austerity. Our mission is to uncover insights on sustainable debt and effective recovery. By examining Greece's experience, we gain a deeper understanding of the dangers of poor fiscal management and the vital role of international organisations. We aspire to inspire discussions on preventing future crises and building resilience in economies worldwide.

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A Journey Through Greece: From Growth to Crisis

The most important foreign policy move which has the biggest impact on Greek domestic life came in May of 1979 when Greece signed the treaty that would make the country a full member of the European Community, beginning in January 1981. In 1981 Greece officially joined the European Union. At that period Greece had a 28% DEBT-GDP ratio, that is the ratio of debt the country has in terms of the GDP. However, in the following years, the two leading political parties of Greece, namely The Panhellenic Socialist Movement known mostly by its acronym PASOK and the New Democracy, who dominated the political stance made the debt-to-GDP ratio go high above 60%(which is considered to be the upper limit for a good economy to maintain their debt-GDP ratio) due to their mismanagement and the debt continued to rise. By 1999 their Debt to GDP ratio went as high as 97%.

In 1981, when Andreas Papandreou[1] became Prime Minister, he and another popular leader shared power. Despite their personal and political differences, things went smoothly. Greece's ties with the Third World and the US, which still mistrusted Papandreou, significantly improved. The Papandreou government, in a policy of National Reconciliation, generously compensated the resistance fighters who fought against the Nazis and the government in the Civil War. It also bravely allowed those who had gone into exile in the Eastern Bloc countries to return.

In 1983, PASOK established the national health system, the Ethniko Systima Ygeias (ESY), which aimed to provide universal healthcare coverage to all Greek citizens and strengthen public health institutions, such as improving the infrastructure of public hospitals and healthcare centres, especially in rural areas, increasing PASOK's support in the countryside. Though Papandreou had been against Greece's entry to the EU, in reality, the agricultural subsidies received by the farmers were an economic boost to the rural population that served to make PASOK more popular.

In 1986 an event outside of Greece caused great anxiety within the country. The nuclear power plant at Chernobyl exploded sending a cloud of radioactivity around the globe and health-threatening levels of radioactive materials were found in at least twenty nations causing much concern in Greece, which was one of the countries outside the former USSR with the highest levels of radioactive contamination. In Greece, they have gotten a taste of the terror of a nuclear world. Scientists expected 50 to 100 deaths from cancer during the next 50 years in Greece because of

[1] Andreas Georgiou Papandreou was a Greek economist, politician, and a dominant figure in Greek politics, known for founding the political party The Panhellenic Socialist Movement, which he led from 1974 to 1996. He served three terms as the 3rd and 8th prime minister in the Third Hellenic Republic.

Chernobyl. Also, the incidence of childhood leukaemia was rising in the more heavily affected parts of Greece due to this accident. While Anti-nuclear sentiment increased all over the world, Greece took a big step when it decided against developing nuclear power.

In 1988, a No-War agreement was signed between Greece and Turkey at Davos, but tensions arose again. That year, a massive fraud scandal occurred which involved the Bank of Crete. Papandreou's enthusiasm turned to bitterness and disbelief as the worst financial and political scandal in four decades engulfed Greece. The press, the Bank of Greece, a magistrate and Parliament were delving into charges of corruption, seeking to uncover how more than $210 million disappeared from the Bank of Crete. This incident shook the country's financial sector.
Additionally, a phone-tapping scandal and allegations of government corruption contributed to PASOK losing popularity.

This marked the growth of 'New Democracy'[2]. Mitsotakis' [3] conservative New Democracy party won the elections without a majority. New elections were called for in April 1988, which they won with a very slim majority.

The collapse of the Berlin Wall in November of 1989 had a significant effect on Greece. Almost instantly, hundreds of thousands of ethnic Greek minorities were sent back to a homeland that many had never visited. Their population was substantial enough to create minority groups in the newly established independent countries, but the rising nationalism in these areas made Greece appear as the most desirable destination for many. The new migrants constituted the most crucial segment of the labour force, taking on the roles that Greeks had abandoned, constructing roads and stone walls, engaging in agricultural and construction work, and working in restaurants and factories. Legal immigrants contributed a larger share of their earnings to social security compared to the Greeks, while utilizing fewer services. The undocumented immigrants accepted lower wages and allowed certain businesses to thrive that might not have survived had they operated legally. Ethnic Greeks who struggle to integrate and lack familial support often fall into the realms of drug use and crime, primarily in larger urban areas.
On January 30th, 1996 a group of Turkish 'journalists' tore down the Greek flag that had been placed on the small island by the mayor of Kalymnos, and replaced it with the Turkish flag. The two countries nearly went to war and finally, after some saber-rattling and serious diplomacy, the crisis was averted. In truth, it was a tricky situation because though according to the International Law of the Sea Treaty, Greece's territorial waters extend 12 miles, some of these small islands are

[2] New Democracy is a liberal-conservative political party in Greece.

[3] Konstantinos Mitsotakis was a Greek politician who was Prime Minister of Greece from 1990 to 1993. His son, Kyriakos Mitsotakis is a Greek politician currently serving as the prime minister of Greece since July 2019,

only a few miles off the coast of Turkey. Despite the incident, relations with Turkey continue to improve with Greece's entry into the European Union and Turkey's desire to do the same.

In 1999 the Athens Stock Exchange looked like a good bet. Greeks who had never even considered putting money into stocks were now investing. Cash that had been concealed in mattresses or concealed in the backyard was being invested in stocks. Novice investors with no knowledge of the market were profiting and informing their friends and family, who also began to invest. Individuals were putting their life savings into the market. Shepherds were selling their sheep to purchase stocks. The market stood at 6000 and was rising. Greek companies such as the ferryboat firm Minoan Lines possess[4] so much wealth that they are unsure how to utilize it. Suddenly the whole bubble burst and just like that all the money was gone. A small number of individuals left early, yet nearly everyone had either lost everything or was aware of someone who had. Fresh in people's minds was the pyramid scheme which had collapsed what was left of Albania's economy a few years before and caused massive riots in that country. But this seemed to be more of a case of collective greed rather than a scheme to rob everyone– the greed of those who pulled their money and got out making billions and the greed of those seduced by the prospect of easy money. Incidentally, the EU launched their EURO currency in 1999, and, on 19 June 2000, the EU, having assessed that Greece fulfils the requirements of the Maastricht Treaty, approved its accession to the euro area as a twelfth member. Greece wanted to be part of that currency. However, to adopt the Euro as their official currency, the EU had set Maastricht criteria, which says that any country willing to use the Euro must fulfil the criteria of a debt-to-GDP ratio to be or under 60%. Surprisingly, Greece adopted the Euro in 2001, which has influenced the Greek economy the most since the 2000s. Greece was only able to gain admission to the EMU(European Monetary Union) by misreporting the size of its deficit, as the country's finances were nowhere near the Maastricht Treaty limits[5]. Greece was hoping that despite its premature entrance, membership in the EMU would boost the economy, allowing the country to deal with its fiscal problems. However, it was not so.

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[4] Passenger ferry company, in Greece.

[5] Goldman Sachs helped Greece hide its debt and appear to have an improving financial position in order to join the Eurozone.

The Impact of the Olympic Mismanagement

In 1992 the Olympic committee had their final round of voting to decide where the 1996 Olympics would be held. Athens was the sentimental favourite and expected to win since after all, it would be the 100th anniversary of the first modern Olympics, which were held in Athens. In Athens everyone expected their city to be named and fireworks were ready to be set off for the inevitable celebration. But in a shocking upset Atlanta comes from behind to win the vote with a public relations barrage and a big pile of money.

The psychological effect of the euro is another story. Suddenly things seem more expensive, perhaps because with drachmas[6] being spent by the thousands even for just a meal, one stops paying attention. In the summer of 2004, Greece finally hosted the Summer Olympics in Athens. As August approached the world was holding its collective breath and wondered if the Greeks could pull it off with construction projects months behind schedule and costs rising astronomically. The American and international media made a point of reporting anything negative having to do with the Olympics with several editorialists questioning Greece's ability and maturity. But as the summer wore on and one major project after another was completed many were forced to concede that the Olympics may actually happen. When Greece stunned the world of soccer by winning the Euro-cup, people were suddenly believers in miracles. For three weeks the entire country watched in amazement as the Greek National Football team knocked off one talented opponent after another. After each victory celebration the feeling was that ”this is great and let's enjoy it while it lasts because next we have to play the French or the Czechs or....” but by the time it had narrowed down to Greece and host Portugal, who had already lost to the Greeks in what was probably the biggest opening game upset in history, everyone in the country knows that when the game ends Greece will be Euro-champions. On the day of the 2004 Athens Olympics opening ceremony the spectators who had chosen to come despite the negative press witnessed the most spectacular opening ceremony in the history of the Olympics and one of the most entertaining games ever, both in the venues and in the sparkling streets of Athens where it seemed every square and plaza had something going on 24 hours a day. The Olympic visitors stayed in newly renovated hotels, rode on the new metro system and the new coastal tram, and walked on miles and miles of newly pedestrianised streets that have turned roads into parks. Greece's stunning run to the European soccer championship and the overwhelming success of the Athens Olympics finished #1 and #2 in

[6] the standard unit of money used in Greece before the introduction of the euro.

an Associated Press poll of the top sports stories of the year. In a CNN poll, the Athens 2004 Olympics were voted the best.

Unfortunately, in the years that followed the games, those in power didn't take advantage of the publicity the Olympics would give them, to promote not only the Athens games themselves but Greece as a tourist destination. Rather than flooding American television with positive images of Greece to counteract the reports of bombs and construction delays, the Greek National Tourist Organization and the Athens Olympic Committee decided to sit back and let CNN do their work for them. Unfortunately, until a week or so before the Olympics, almost all the news was bad, which was of course the nature of the news media. It was too late when suddenly people had second thoughts about skipping these Olympics and decided not to come. So while the Olympics were a success, the empty seats put an asterisk on the word successful. There were no terrorist attacks and everybody there had a great time, but it was a missed opportunity and the total bill for the games was a staggering 15 billion euros and counting. The Olympics compelled the government to make changes to the infrastructure that needed to be made and it was doubtful that all of them would have been done without the Olympics. When the whole world is watching you, then you follow through on your responsibilities. On the other hand, there were lost opportunities and lost money from construction delays and corruption. They hoped that an Olympic hangover would not take too much of a toll on the economy of the country.

The Hope

The period of rule by New Democracy, led by Costa Karamanlis, is comparable to the George W Bush years in the United States with jobs and positions awarded to loyalists, family and friends instead of people with ability and experience. Several major scandals threatened to topple the government but each one was quashed by parliamentary manoeuvring and cover-ups and few perpetrators (if any) were punished. In the meantime, the frustration of the Greek people grew and the smaller parties began to take away votes from the two major parties. The Sieman's scandal[7] features millions of euros in bribes over telecommunication contracts. The Vatopedi Monastery Scandal features a land trade, valuable property owned by the government for useless land owned by a wealthy monastery on Mount Athos that again cost the Greek people millions. Other scandals include a shipping company bribing a minister for subsidized ferry routes, overpaying for stocks and bonds for pension funds, broken agreements with workers, and a million other things. While ND and PASOK hurled insults at each other, the country was falling apart. Greece experienced a series of devastating forest fires in August 2007. The fires burned over 180,000 hectares of forest and agricultural land. Once again the nation’s international recognition went wrong.

But it was in December of 2008 that the New Democracy government showed how powerless it was when a policeman in the student neighbourhood of Exarchia shot a 15-year-old boy, this set off a wave of violence as the clubs, which were full on Saturday night, empty out as young people burn and smash any building that houses a business that even appears to be part of the global conspiracy. The police stood by and watched, hoping to contain the violence and not instigate more. But the people of Athens saw it differently, that the government was not able to protect them. Some people took advantage of the violence to loot, others to settle scores with their neighbours and competitors and use the violence to serve their own purposes. The student who was killed, Alexander Grigoropoulou, was elevated to martyrdom by some of the media and the events were used by political parties to further their objectives, some to the point of encouraging the violence. But in the end, the 'riots' peter out and life in Athens goes back to normal, meaning daily demonstrations and occasional skirmishes between anarchists and the police instead of battles, and the smashing of ATMs instead of the burning of banks.

[7] The Siemens bribery scandal in Greece is a corruption and bribery scandal in Greece over deals between Siemens and Greek government officials during the 2004 Summer Olympic Games in Athens, Greece regarding security systems and purchases by OTE in the 1990s

Greek Financial Crisis and Bailout

The global financial crisis that began in 2007 exposed the true nature of Greece's financial strife. The recession weakened Greece's already paltry tax revenues, which caused the deficit to worsen. Furthermore, when their government changed in 2009, the new Prime Minister[8] revealed that the previous government had been fudging data on a large scale for years. They were saying that their budget deficit was 6.7%, but it was crossing 15%. This maligned Greece's reputation across the world. Investors lost their trust and felt they could not trust this country and should withdraw their investments. This country's credit rating was downgraded, and people were scared of investing their money because they knew that the country could not afford to pay back. So overall, until 2010, the debts continued to rise and no one was ready to lend any money. The economy began to crash drastically.

In 2010, U.S. financial rating agencies stamped Greek bonds with a "junk" grade. Greece found itself at the centre of a wave of criticism by the international press, international organizations, rating agencies and the European Commission. Despite the fact that the fiscal situation in 2009 worsened throughout Europe and the rest of the world, in many countries much more than in Greece, Greece was the first sovereign to find itself in the middle of a confidence crisis. There are three likely reasons for this.

- The first was the high level of Greece's public debt. Greece's public debt had stabilized since the early 1990s at roughly 100% of GDP, versus 70% for the average of the Euro area.
- The second reason was the announcement of a large deterioration of the projected deficit for 2009, by the government elected in October 2009. This took the markets by surprise and contributed to the confidence crisis.
- The third reason is related to the shortcomings of the fiscal program initially adopted by the newly elected government, which appeared to be leading to a further widening of the fiscal deficit rather than a contraction.

As capital began to dry up, Greece faced a liquidity crisis, forcing the government to seek bailout funding, which they eventually received with staunch conditions.
Bailouts totalling €289 billion from the International Monetary Fund (IMF) and other European creditors were conditional on Greek budget reforms, specifically, spending cuts, higher tax

[8] George Andreas Papandreou is an American-born Greek politician who served as Prime Minister of Greece from 2009 to 2011. He is currently serving as an MP for Movement for Change.

revenues and even privatisation of government assets. These austerity measures created a vicious cycle of recession with unemployment reaching 25.7% in August 2012.

In the following pages, we aim to explain the narrative of Greece's debt crisis, emphasizing the importance of exploring the **Theory of Debt Sustainability** for a deeper understanding. This framework analyzes the root causes of unsustainable debt, which include high public debt, negative economic growth, rising borrowing costs, and chronic fiscal deficits. These factors undermine fiscal stability and investor confidence. Our study employs a clear mathematical and econometric approach.

The Theory of Debt Sustainability

Debt sustainability, which is a crucial concept in public finance and macroeconomic analysis, examines the relationship between a nation's fiscal policy, economic growth, and debt dynamics. It is crucial for assessing Greece's economic crisis, as it helps evaluate the root causes, effectiveness of policy measures, and recovery strategies.

The Greek economic crisis emerged because public debt became unsustainable, leading to a loss of investor confidence and forcing Greece out of the financial markets.
Debt sustainability theory helps explain why this happened by examining the interaction of:

- High debt levels
- Low or negative GDP growth
- Rising borrowing costs
- Chronic fiscal deficits

Without this theoretical perspective, comprehending the ongoing increase in debt amidst Greece's attempts to stabilise its economy becomes challenging.

Assessing Policy Effectiveness:

Debt sustainability theory offers a structured way to evaluate how different policy measures affect the debt trajectory:

Austerity Measures:

- How do reductions in spending and increases in taxes (f_t) affect the debt-to-GDP ratio? Are these actions adequate to counterbalance the adverse effects of a declining GDP?

Growth Policies:

- What role does economic growth (g_t) play in stabilizing or reducing debt? Can structural reforms effectively boost growth?

Debt Restructuring:

- How does lowering interest rates (r_t) or extending maturities improve sustainability?

Using the following framework (**a simplified/narrower version of the actual solvency model, we created a two-period mathematical model)**, policymakers can weigh the trade-offs of various options and understand why Greece's austerity measures alone were insufficient without accompanying growth policies.

Derivation of the model:

- The debt-to-GDP ratio at time $(t+1)$ is given by:

$$\frac{d_{t+1}}{y_{t+1}} = \frac{debt\ at\ time\ period\ (t+1)}{gdp\ at\ time\ period\ (t+1)}$$

- The debt at time t+1 consists of:
  - Existing debt at period t (d_t), is increased by the interest rate (r_t).
  - Primary fiscal deficit (f_t).
- Thus

$$d_{t+1} = (1+r_t)d_t + f_t$$

- GDP grows over time at a rate g_t . Therefore:

$$y_{t+1} = (1+g_t)y_t$$

- So substituting we get:

$$\frac{d_{t+1}}{y_{t+1}} = \frac{(1+r_t)d_t}{(1+g_t)y_t} + \frac{f_t}{(1+g_t)y_t}$$

Economic Intuition:

Interest-Growth Differential ($r_t - g_t$):

- If the economy grows faster than the interest rate ($g_t > r_t$), the debt-to-GDP ratio can decrease even with a primary deficit.
- Managing debt becomes increasingly difficult if ($r_t > g_t$).

Primary Deficit (f_t):

- A higher deficit increases the debt-to-GDP ratio.

The Four Factors that Affect Fiscal Sustainability and Solvency

Factors determine whether a government is solvent:

- the predetermined (historical) debt-to-GDP ratio,
- the real interest rate on government bonds,
- the growth rate of real GDP,
- the primary surplus as a share of GDP.

- ➢ The historical sovereign debt to GDP is predetermined and cannot be changed.
- ➢ With full capital mobility, for a small open economy in a monetary union, such as Greece's, the real interest rate is also largely exogenous and outside the immediate control of the government.
- ➢ In an open economy, both in the short term and medium term, the growth rate of GDP depends on many factors(explained in the following econometric model). All those factors can be affected by government policy only indirectly. In the long run, the growth rate may well be exogenous, determined by population growth and technological progress.

Thus, the primary element under the direct control of a government in both the short run and the medium run is the ratio of the government's primary surplus to GDP. This metric serves as the principal tool for a government to achieve fiscal solvency.

Before analyzing the government's implemented policies, we established our econometric model, which enabled us to work with empirical data rather than rely solely on theoretical frameworks.

Econometric Model for Analysis

Summary of the Model Results:

To examine the macroeconomic determinants of Greece's Government Debt-to-GDP Ratio, we employ a Vector Autoregression with Exogenous Variables (VARX(5)) model.

VAR Order Selection (* highlights the minimums)

	AIC	BIC	FPE	HQIC
0	14.48	14.74	1.947e+06	14.57
1	5.193	6.966	183.9	5.834
2	2.567	5.860	15.08	3.758
3	1.799	6.612	9.942	3.539
4	-0.4503	5.883	2.334	1.840
5	-7.510*	0.3432*	0.01207*	-4.671*

This methodological approach is well-suited for capturing the dynamic interactions among multiple economic indicators, allowing us to analyze how debt levels respond to changes in GDP growth, unemployment, government spending, tax revenues, and foreign direct investment (FDI) over time. Additionally, we account for Greece's participation in the Eurozone and the European

Union, recognizing their potential influence on fiscal dynamics. By incorporating GDP Growth, Unemployment**,** Government Spending**,** Tax Revenues**,** FDI, and EU-related factors, the model seeks to provide insights into the contributing factors to the country's rising debt levels.

The results highlight several key points:

- GDP Growth shows mixed effects over different time lags, indicating that economic growth plays a crucial yet delayed role in influencing debt dynamics.
- Unemployment has a significant negative effect on debt, implying that higher unemployment can increase government debt through rising expenditures and reduced tax revenues.
- Government Spending and Tax Revenues have intricate and varying effects on the debt ratio, suggesting that fiscal policies are not always immediate in their impact and depend on the timing and magnitude of government interventions.
- FDI exhibits a generally negative relationship with government debt, possibly reflecting external challenges that hinder the effectiveness of foreign investment.

- The EU-related factors (Eurozone membership) significantly contribute to the debt trajectory, with the model suggesting that Greece's position within the EU has both positive and negative effects on its fiscal health.

Results for equation Government Debt-to-GDP Ratio

	coef	std err	z	P>|z|	[0.025	0.975]
intercept	33.9018	0.001	2.72e+04	0.000	33.899	33.904
L1.GDP Growth (%)	-0.2509	0.022	-11.380	0.000	-0.294	-0.208
L1.Unemployment Rate	0.7329	0.016	46.783	0.000	0.702	0.764
L1.Government Debt-to-GDP Ratio	0.3946	0.055	7.171	0.000	0.287	0.502
L1.Government Spending to GDP (%)	-2.5014	0.045	-55.055	0.000	-2.590	-2.412
L1.Tax Revenues (% of GDP)	-2.4015	0.022	-111.661	0.000	-2.444	-2.359
L1.FDI Amount (USD Billion)	-1.7569	0.007	-261.128	0.000	-1.770	-1.744
L2.GDP Growth (%)	0.1584	0.045	3.513	0.000	0.070	0.247
L2.Unemployment Rate	-0.1875	0.013	-14.124	0.000	-0.214	-0.161
L2.Government Debt-to-GDP Ratio	0.3736	0.069	5.405	0.000	0.238	0.509
L2.Government Spending to GDP (%)	-0.7797	0.046	-16.805	0.000	-0.871	-0.689
L2.Tax Revenues (% of GDP)	-0.9263	0.021	-44.274	0.000	-0.967	-0.885
L2.FDI Amount (USD Billion)	-1.7916	0.007	-254.560	0.000	-1.805	-1.778
L3.GDP Growth (%)	-0.4662	0.037	-12.485	0.000	-0.539	-0.393
L3.Unemployment Rate	-0.4727	0.015	-32.160	0.000	-0.502	-0.444
L3.Government Debt-to-GDP Ratio	0.2150	0.052	4.150	0.000	0.113	0.317
L3.Government Spending to GDP (%)	0.0737	0.048	1.540	0.124	-0.020	0.168
L3.Tax Revenues (% of GDP)	-0.1269	0.016	-7.834	0.000	-0.159	-0.095
L3.FDI Amount (USD Billion)	-1.1360	0.007	-153.746	0.000	-1.151	-1.122
L4.GDP Growth (%)	-0.6888	0.033	-21.092	0.000	-0.753	-0.625
L4.Unemployment Rate	-0.7070	0.021	-33.727	0.000	-0.748	-0.666
L4.Government Debt-to-GDP Ratio	0.0028	0.083	0.034	0.973	-0.160	0.165
L4.Government Spending to GDP (%)	1.2551	0.043	29.359	0.000	1.171	1.339
L4.Tax Revenues (% of GDP)	0.1676	0.019	8.736	0.000	0.130	0.205
L4.FDI Amount (USD Billion)	-0.7468	0.007	-112.722	0.000	-0.760	-0.734
L5.GDP Growth (%)	-0.5395	0.042	-12.900	0.000	-0.621	-0.458
L5.Unemployment Rate	-1.0195	0.012	-82.112	0.000	-1.044	-0.995
L5.Government Debt-to-GDP Ratio	-0.2022	0.089	-2.264	0.024	-0.377	-0.027
L5.Government Spending to GDP (%)	2.4577	0.044	55.385	0.000	2.371	2.545
L5.Tax Revenues (% of GDP)	0.1125	0.022	5.133	0.000	0.070	0.155
L5.FDI Amount (USD Billion)	-0.7256	0.009	-84.615	0.000	-0.742	-0.709
beta.EUROZONE(EURO as the official currency)	59.8802	0.004	1.53e+04	0.000	59.873	59.888
beta.EU Member	51.6868	0.001	4.15e+04	0.000	51.684	51.689

While the model provides important insights, it also has certain limitations, including issues related to multicollinearity**,** autocorrelation**,** and heteroskedasticity, which may affect the reliability of the results. These limitations highlight the need for further refinement of the model, potentially through techniques like robust standard errors, addressing endogeneity, and optimizing the lag length selection to ensure a more accurate representation of the complex relationship between the variables.

In conclusion, this model provides a foundational understanding of the key economic factors influencing Greece's debt dynamics during the crisis. However, addressing its limitations will be essential for drawing more definitive conclusions and ensuring the model's robustness for future policy analysis.

The provided regression model investigates the relationship between the Government Debt-to-GDP Ratio and key macroeconomic indicators, focusing on the Greek financial crisis. By

including various economic variables such as GDP Growth**,** Unemployment**,** Government Spending**,** Tax Revenues**,** FDI, and EU-related factors, the model aims to understand how these factors interact over time and contribute to the changes in Greece's government debt levels.

The general regression equation presented is as follows:

$$
\begin{aligned}
\text{GovDebt}_t = {} & 33.9018 \\
& - 0.2509 \cdot \text{GDPGrowth}_{t-1} + 0.1584 \cdot \text{GDPGrowth}_{t-2} - 0.4662 \cdot \text{GDPGrowth}_{t-3} \\
& - 0.6888 \cdot \text{GDPGrowth}_{t-4} - 0.5395 \cdot \text{GDPGrowth}_{t-5} \\
& + 0.7329 \cdot \text{Unemployment}_{t-1} - 0.1875 \cdot \text{Unemployment}_{t-2} - 0.4727 \cdot \text{Unemployment}_{t-3} \\
& - 0.7070 \cdot \text{Unemployment}_{t-4} - 1.0195 \cdot \text{Unemployment}_{t-5} \\
& - 2.5014 \cdot \text{GovSpending}_{t-1} - 0.7797 \cdot \text{GovSpending}_{t-2} + 1.2551 \cdot \text{GovSpending}_{t-4} \\
& + 2.4577 \cdot \text{GovSpending}_{t-5} \\
& - 2.4015 \cdot \text{TaxRevenues}_{t-1} - 0.9263 \cdot \text{TaxRevenues}_{t-2} - 0.1269 \cdot \text{TaxRevenues}_{t-3} \\
& + 0.1676 \cdot \text{TaxRevenues}_{t-4} + 0.1125 \cdot \text{TaxRevenues}_{t-5} \\
& - 1.7569 \cdot \text{FDI}_{t-1} - 1.7916 \cdot \text{FDI}_{t-2} - 1.1360 \cdot \text{FDI}_{t-3} \\
& - 0.7468 \cdot \text{FDI}_{t-4} - 0.7256 \cdot \text{FDI}_{t-5} \\
& + 59.8802 \cdot \text{EUROZONE} + 51.6868 \cdot \text{EU Member} + \varepsilon_t
\end{aligned}
$$

This equation suggests that the Government Debt-to-GDP Ratio in Greece is influenced by a complex interplay of past economic conditions. Key insights from the model include:

1. Lagged Effects of GDP Growth: The coefficients for past values of GDP Growth show a mixed impact on the government debt ratio, with both positive and negative influences observed at different time lags. This indicates that the effects of economic growth on debt accumulation may take time to materialize and can vary depending on the state of the economy in previous periods.
2. Impact of Unemployment: Unemployment plays a crucial role in shaping Greece's debt-to-GDP ratio, with significant negative coefficients observed across multiple lags. This suggests that higher unemployment leads to reduced tax revenues and potentially higher government spending (such as unemployment benefits), contributing to rising debt levels.
3. Government Spending and Tax Revenues: The model highlights the intricate relationship between Government Spending and Tax Revenues with the debt ratio. While some lags of Government Spending appear to increase the debt ratio, others show a negative relationship, emphasizing that the effects of fiscal policies are not immediate and depend on the timing and extent of government interventions.
4. FDI's Role: The negative coefficients for FDI across various lags suggest that foreign direct investment may not have had the desired effect in reducing government debt during the Greek crisis. This could be due to external factors like global economic uncertainty or structural weaknesses in the domestic economy that hinder the beneficial impact of FDI.

5. EU-Related Variables: The inclusion of the EUROZONE and EU Member variables suggests that Greece's membership in the European Union, especially the Eurozone, has had a significant impact on its debt trajectory. The positive coefficients for these variables may indicate the influence of fiscal and monetary policies implemented at the EU level, which could have affected Greece's ability to manage its debt.

While the model offers valuable insights, several limitations need to be addressed to enhance its robustness and interpretability:

```
Covariance Type:
==========================================================================================================================
Ljung-Box (L1) (Q):     1.38, 18.55, 21.10, 12.90, 6.55, 11.61   Jarque-Bera (JB):      3.67, 6.88, 6.04, 0.33, 42.11, 0.22
Prob(Q):                  0.24, 0.00, 0.00, 0.00, 0.01, 0.00   Prob(JB):               0.16, 0.03, 0.05, 0.85, 0.00, 0.90
Heteroskedasticity (H):    3.12, 2.77, 0.25, 2.54, 13.51, 0.71   Skew:               -0.16, 0.88, 0.85, -0.16, -1.12, -0.02
Prob(H) (two-sided):       0.03, 0.06, 0.01, 0.08, 0.00, 0.52   Kurtosis:               4.36, 3.77, 3.59, 2.72, 7.18, 3.34
```

1. Multicollinearity: The model includes several potentially correlated variables (e.g., Government Spending**,** Tax Revenues, and GDP Growth), which could lead to multicollinearity. This might inflate standard errors and obscure the true relationship between the predictors and the dependent variable. To address this, further diagnostics like Variance Inflation Factor (VIF) should be performed, and highly correlated variables could be reconsidered.
2. Autocorrelation: Evidence from the Ljung-Box test suggests potential autocorrelation in the residuals, which violates the assumption of independent errors. This could lead to biased parameter estimates and incorrect statistical inference. Addressing this by using robust standard errors or models that account for autocorrelation (e.g., ARMA) is recommended.
3. Heteroskedasticity: The Heteroskedasticity test indicates that the variance of the residuals changes over time. This means the model may not account for the varying uncertainty across different levels of the independent variables. Using heteroskedasticity-robust standard error**s** or adopting a Generalized Least Squares (GLS) approach would be beneficial.
4. Endogeneity: There may be concerns about endogeneity, especially with variables like Government Spending and Tax Revenues, which could be affected by the Government Debt-to-GDP Ratio itself. To address this, techniques like Instrumental Variables (IV) or Two-Stage Least Squares (2SLS) could be used.
5. Residual Normality: The Jarque-Bera test indicates that the residuals are not normally distributed, which may affect the accuracy of statistical inference. This could be addressed by transforming the dependent variable or exploring alternative distributions.

```
No. Observations:                   45
Log Likelihood                -584.321
AIC                           1606.641
BIC                           2002.301
HQIC                          1754.139
```

6. Lag Length Selection: The model uses 5 lags for each variable, but this choice was not explicitly tested. To improve the model, the optimal lag length should be determined using criteria such as AIC, BIC, or HQIC.

This regression model sheds light on the complex dynamics behind the evolution of Greece's Government Debt-to-GDP Ratio during the crisis, highlighting the interplay between fiscal policies, economic growth, unemployment, and external factors like EU membership and foreign investment. However, to ensure robustness and reliability, addressing issues like multicollinearity, autocorrelation, and endogeneity, as well as optimizing lag lengths, will be crucial in refining the model and making it more reflective of the underlying economic processes.

Expectations: The Invisible Power Driving the Greek Debt Crisis

We often talk about a country's solvency i.e., its ability to pay its debts. That's a crucial piece of the puzzle, but it’s not the whole story. Even if a country could theoretically repay, a crisis can erupt if people start to doubt its will. This is where expectations come in, they are the driving force in financial markets, and that’s what happened in Greece. Consider the bond market, where governments borrow money, as a marketplace where trust is the most valuable currency. When people trust a government to repay its debts, everything works smoothly. But, when that trust fades, things can unravel very quickly. As the marketplace is driven by what investors believe will happen, if everyone believed the Greek government would pay them back, they would be happy to hold onto Greek bonds, because they would be confident they would get their money back, and they would accept lower interest rates. This would keep borrowing costs manageable for Greece.

But what if doubt creeps in? Suddenly, people are asking, "What if Greece can't pay us back?". Then that expectation becomes a self-fulfilling prophecy. It's like a rumour spreading through the market: the more people who start thinking about the possibility of default, the more others start feeling the same. This is how a market can reach a point where it perceives only the possibility of default. Think of it like a crowded theatre. If someone shouts "fire" and people believe it, everyone rushes towards the exit which can easily create a dangerous situation, regardless of whether there is a fire or not. Similarly, if people fear the Greek default, they will sell Greek bonds in a rush, which will drastically increase the interest rates and it will become impossible to get new loans, which leads to default.

So the first question that hits in mind is "Why is it that the same markets that comfortably held and continuously refinanced the debt of a sovereign state for years on end, suddenly lose faith and refuse to touch it?" To answer this question, one can start with the equilibrium relationship between the return of domestic bonds (deemed unsafe) and a safe bond. With risk-neutral investors, in equilibrium, the relationship between the real interest rate of a domestic (Greek) bond and a safe (German) bond, would be given by,

- $(1-\pi)(1+r) = 1 + r*$

- π is the expected probability of default of the domestic bond.
- r the domestic real interest rate.
- r* the safe interest rate.

From the equation, the probability of default is given by,

- $\pi = (r - r*\ )/(1 + r)$

This gives us the equilibrium relation between the perceived probability of default of the domestic bond π, and the spread between the interest rates of the domestic and the safe bond $(r - r*)$. The two are positively related: If the perceived probability of default is zero, then the domestic interest rate will be equal to the interest rate on the safe bond. As the probability of default increases, so does the spread and the domestic interest rate. As the probability of default approaches unity (100%), the spread, and the domestic interest rate approach infinity. The more people worry, the higher the cost of borrowing becomes for Greece, and it can make default more likely.

There are several characteristics of these models, which are relevant to the Greek crisis.

- First, small changes in the fundamentals may cause an economy to shift from a low-spread equilibrium to a high-spread equilibrium, or even to default.
- Second, such a change in the nature of the equilibrium is always unanticipated. In fact, it occurs when expectations suddenly change.
- Third, a change in the nature of the equilibrium can take place even without changes in the fundamentals, just because the assessment of some investors about the probability of default changed, and these investors are not willing to hold the bonds of the country in question.

These will force even investors, who have not changed their beliefs, to change their behaviour as they may become worried about whether there will be enough investors in the future willing to hold the bonds in question.

The fundamentals of the Greek fiscal situation changed because of the **2008 international financial crisis** and **the recession** that ensued, causing investors to question the ability of Greece to keep servicing its debt. This, in turn, brought about a liquidity crisis in the market for Greek bonds, and since there were few signs that Greece was prepared to adjust its primary surplus to

achieve solvency, and no committed lender of last resort, Greece was cut off from the international bond market. It avoided an outright default because it could turn to a special European support mechanism that was devised in a rush.

Now, for years, investors had been comfortable with Greek debt. While Greece had its issues before, people generally thought things would be okay. There were indeed persistent issues with public finances and international competitiveness. However, the international financial crisis of 2008 shook everything up, and those issues were brought to light. This made the markets reconsider and realize that Greece's high debt level meant it was in a vulnerable situation. This crisis made them reconsider, revealing hidden instability, similar to a house with a crack showing a long-standing weakness, raising doubts about their optimism.

The government of Greece did not contribute positively to the situation. The markets received the news that the fiscal deficit had exploded to 13.6% of GDP in 2009. This shift was significant because it moved away from a state of managed deficits to an unmanageable situation. Then, the newly elected government came clean about the actual state of their finances which was a lot worse than what was previously believed. But that wasn't all, the new government didn't take credible steps to show that they were serious about the deficit. It was all statements, no real action. In fact, in its first few months in office, it further contributed to the deterioration of the fiscal situation and the budget for 2010 was not deemed credible by the markets. This was a major turning point. The revelation of worse-than-expected fiscal data, coupled with the new government's actions, destroyed whatever lingering faith investors had. They began to see Greece as an entity that was neither able nor willing to address its financial woes. The lack of a credible budget plan intensified fears, prompting investors to re-evaluate and start selling bonds.

This led to a **crisis of confidence** and was like pouring fuel on the fire. The market’s fear started to grow, which meant that the interest rates on Greek bonds started to jump dramatically. Spreads, which is the difference between Greek bonds and safe (e.g., German) bonds, went from 30 basis points at the end of 2007 to 200 basis points at the end of 2008, and this trend went further. This indicated a distinct change in the market’s evaluation of creditworthiness. Everyone was watching the widening gap. Rating agencies, which play a role in the marketplace's expectations, started to downgrade Greece's debt which made things worse. The markets were not convinced by statements from the Eurogroup of support for Greece either.

Essentially, Greece found itself in a position where it couldn't get new loans at a sustainable rate. The market simply didn't believe Greece could manage its debt anymore and was effectively forcing its hand into a default. While Greece had a history of spending problems and a long list of

missed reforms, at the core of its debt crisis was the collapse of expectations. It further fuelled the solvency crisis, which resulted in a crisis of confidence. It wasn't just about the debt itself, but about a widespread loss of faith in Greece's ability to fix things.

The **strict austerity measures**, applied amidst the worst financial crisis since the Great Depression, proved to be one of the largest factors attributing to Greece's economic implosion. Tax Revenues weakened, which made Greece's fiscal position worse. These measures also created a humanitarian crisis: homelessness increased, suicides hit record highs(considered in the econometric model **(chart 9 in appendix)**), and public health significantly deteriorated.

Greek citizens voted against additional EU austerity measures in July 2015. Afterwards, Greece began to recognize improvements in its economy. The nation's unemployment rate went from a record high of 28% in 2014 to 13.2% in 2021.
The country's GDP has also been wildly volatile. For instance, Greece's annual year-over-year GDP growth went from -10.1% in 2010 to 1.8% in 2019. Unfortunately, it has been inconsistent since then. Greece's year-over-year GDP growth was -9% in 2020, 8.4% in 2021, and 5.9% in 2022.

Restrictions on Exploring Other Solvency Channels Which Made Greece Suffer

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In Greece's case during the Eurozone crisis, there were alternative solvency channels that could have been explored or emphasized more effectively by the Greece government which included:

- **Debt Restructuring :**

  This simply means reducing the debt burden through restructuring. Early in the crisis, there was reluctance among creditors and EU institutions to consider debt restructuring or write-downs. This delayed Greece's recovery and exacerbated its debt burden.

- **Fiscal Transfers or Grants:**

  Lack of Financial Support from wealthier Eurozone members in the form of direct grants, not just loans. Grants would have reduced the reliance on loans with punitive conditions and helped fund growth initiatives. However, the Eurozone lacked sufficient fiscal transfer mechanisms to support struggling economies like Greece during crises.

- **Monetary Policy Constraints:**

  As a member of the Eurozone, Greece lacked control over its monetary policy. It couldn't devalue its currency to make its exports more competitive or print money to stimulate the economy. This limitation left Greece heavily dependent on external loans and forced to adhere to strict fiscal discipline.

- **Structural Reforms:**

  Reforms in labour markets, public administration, and business regulations improve productivity and competitiveness but the economy of Greece was already vulnerable due to structural issues such as tax evasion, inefficiency in public administration, and reliance on borrowing. Hence, even if some structural reforms were attempted, they were often implemented under external pressure and lacked coordination, undermining their effectiveness.

- **Privatization with Strategic Planning:**

This means selling state-owned assets to raise funds while ensuring long-term benefits for the economy. Greece implemented some privatizations, but they often faced criticism for being poorly planned, rushed, and undervalued. Hence, failed to help the economy of Greece.

Hence these restrictive policies, combined with limited flexibility in adopting alternative solvency mechanisms, worsened Greece's economic crisis, particularly during the Eurozone debt crisis.

Conclusion

The Greek economic crisis serves as a profound case study of the complexities of fiscal management, debt sustainability, and the interplay of domestic policies and global economic forces. The crisis not only exposed Greece's structural weaknesses and fiscal mismanagement but also underscored the critical role of investor confidence, effective governance, and international cooperation in navigating economic downturns.

Our analysis, supported by theoretical and econometric models, highlights the multifaceted nature of debt sustainability. It reveals how high debt levels, stagnant economic growth, rising borrowing costs, and chronic fiscal deficits interacted to create a vicious cycle of recession and financial instability in Greece. The austerity measures implemented during the bailout programs, while addressing immediate fiscal imbalances, further deepened economic contractions, demonstrating the need for a balanced approach that incorporates growth-oriented policies alongside fiscal discipline.

Looking ahead, the lessons from Greece's crisis underline the importance of proactive governance, transparent fiscal policies, and resilient economic frameworks. For countries facing similar challenges, the adoption of robust debt sustainability frameworks, complemented by structural reforms and international support, is essential to mitigate risks and foster economic recovery.

Appendix

Chart 1

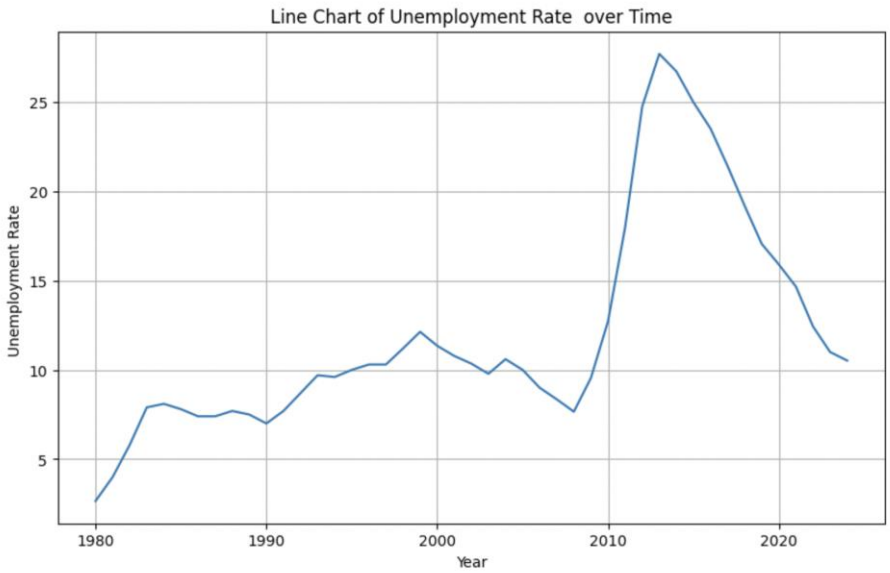


Chart 2

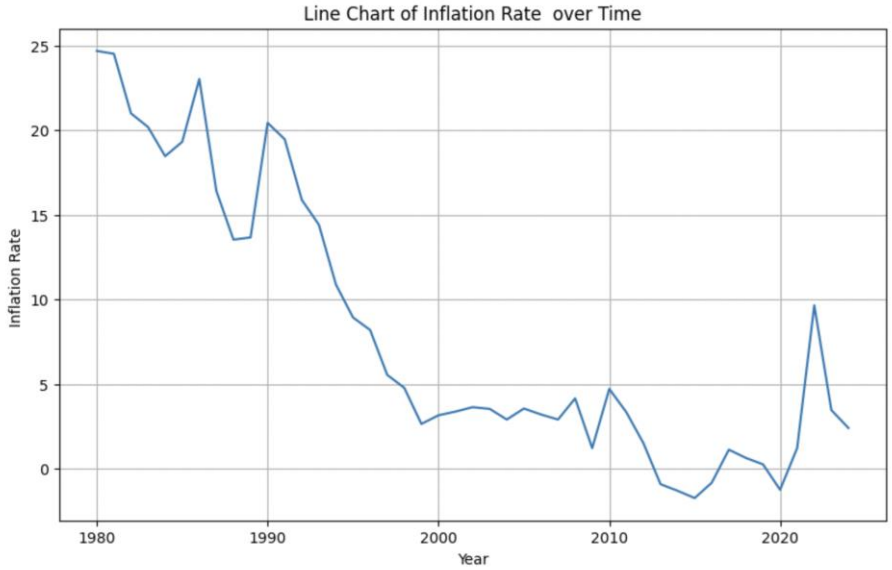
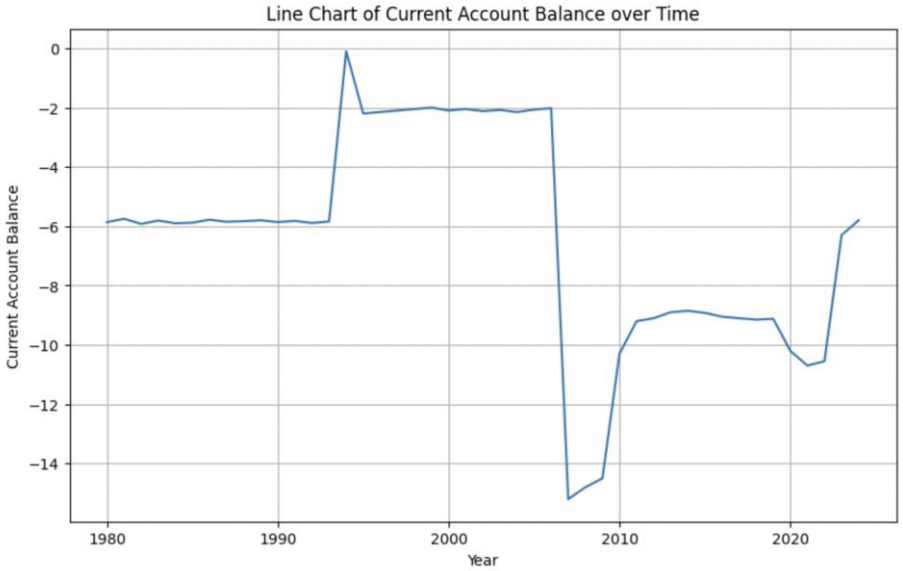


Chart 3



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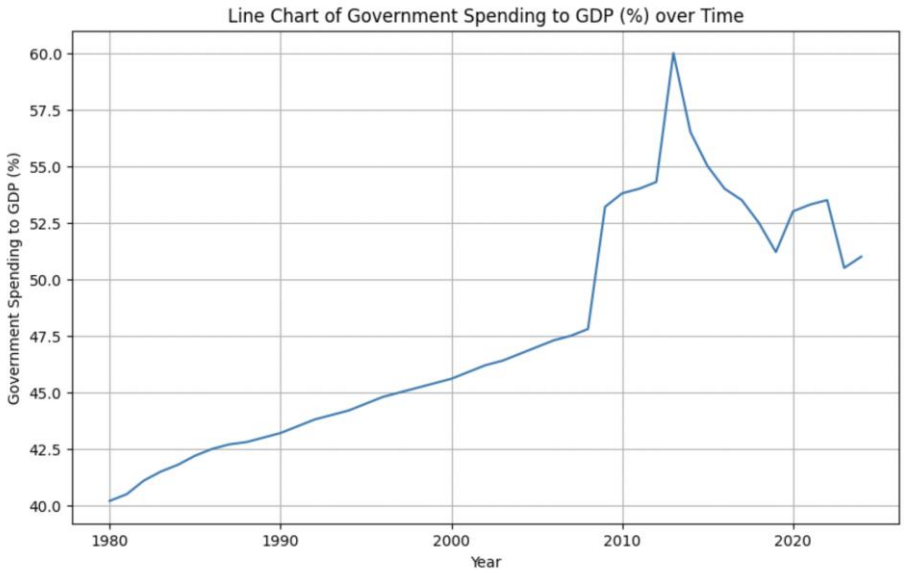

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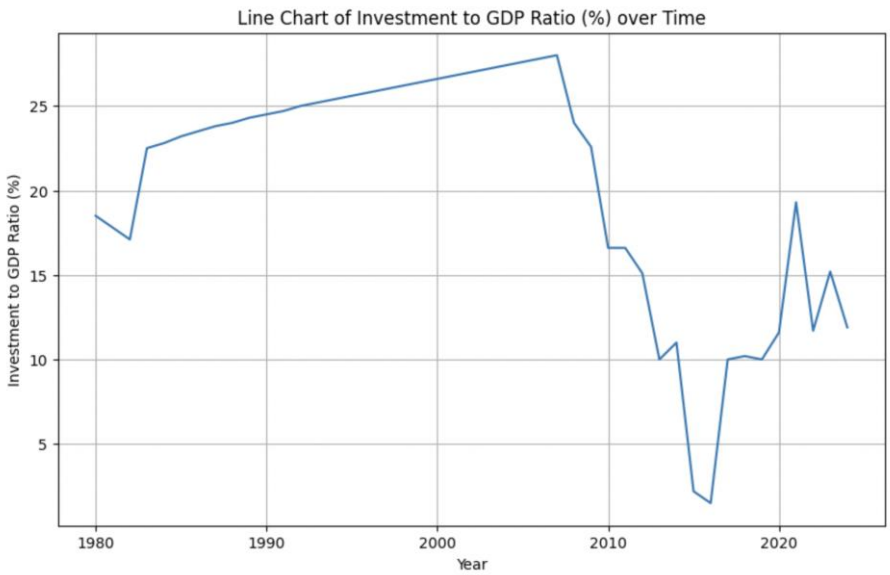

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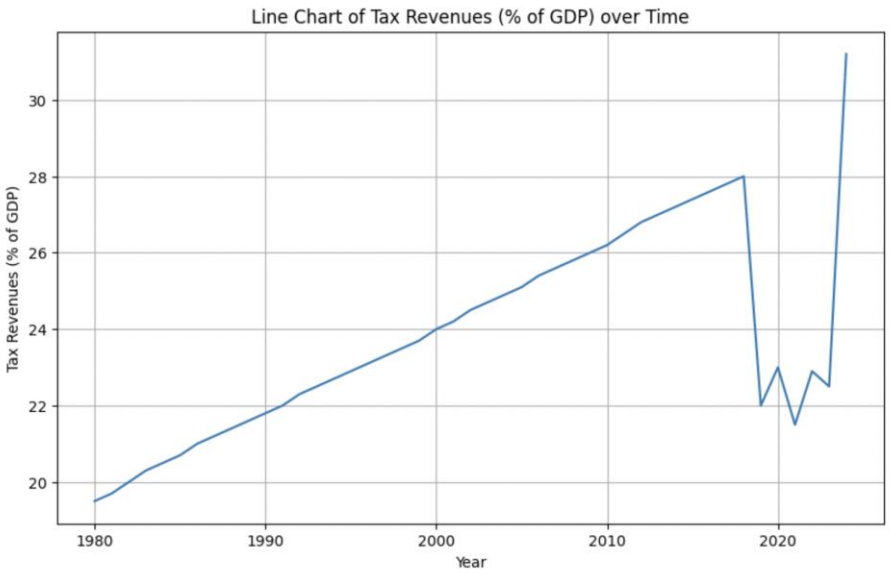

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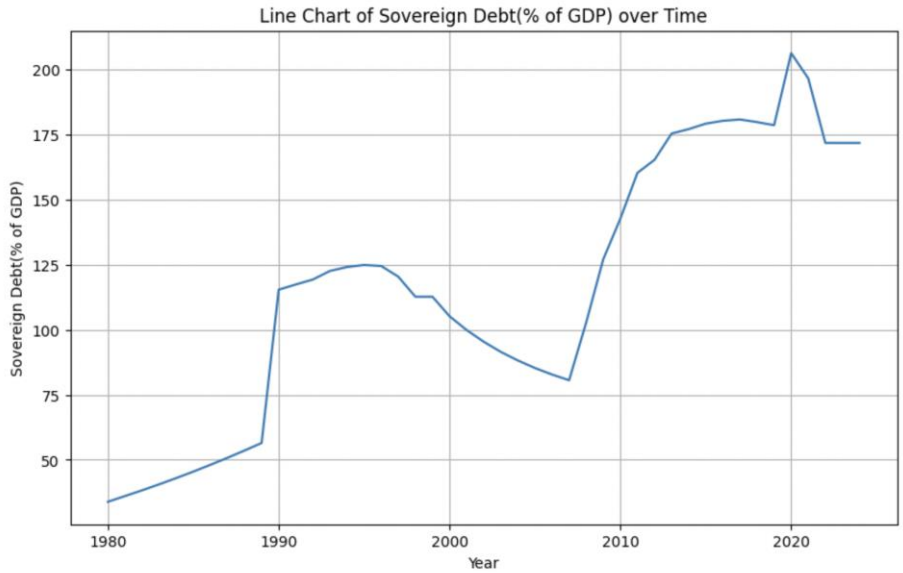

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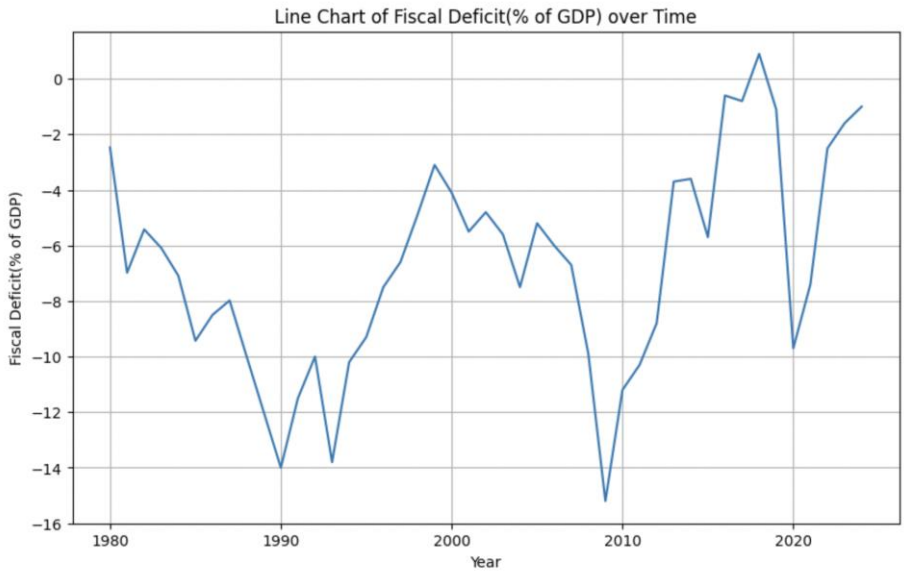
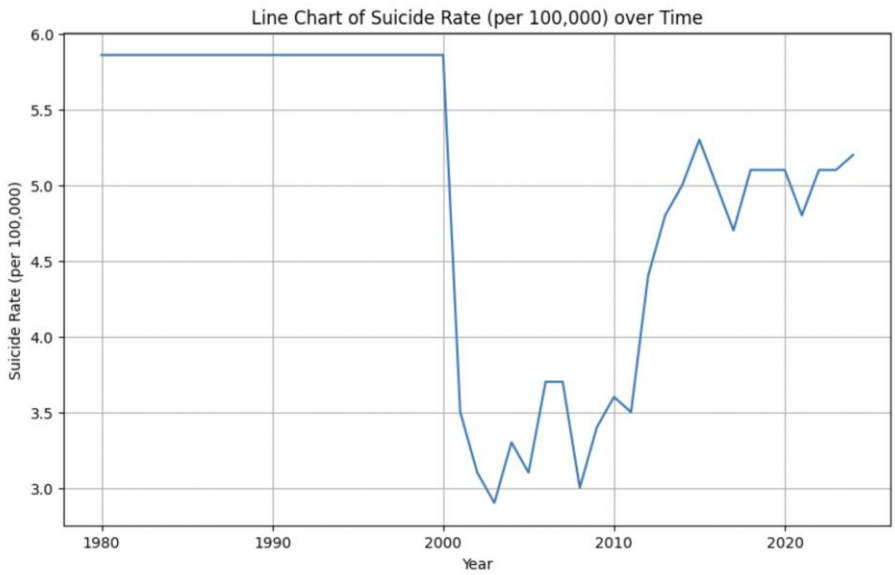


Chart 9


Line Chart of Suicide Rate (per 100,000) over Time
Suicide Rate (per 100,000)
6.0
5.5
5.0
4.5
4.0
3.5
3.0
1980
1990
2000
2010
2020
Year

Bibliography

- https://colab.research.google.com/drive/1N-kufDSdWbgOOAYda_uP-zOP4H_GpN0V#scrollTo=QPZeFke96Q1F (Econometric analysis)
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